American Terawatt

TEMPLATE TEST

# Cable Link Single-Line Diagram

## TikZ rendering of the sending-end / receiving-end transmission model

Test document for the shared report template. It compiles a TikZ rendering of the single-line diagram of a cable transmission link (source, disconnectors, cable of length $L_{\mathrm{cable}}$ and load impedance $Z_L$) with the sending-end and receiving-end voltages, currents and power flows.

## Revision history

| REV. | DATE | AUTHOR | DESCRIPTION |
|---|---|---|---|
| 0.1 | Sep. 18, 2026 | Eugen Starschich | Initial document |

# 1 Transmission model

Figure 1 shows the single-line representation of a cable transmission link between a sending end $S$ and a receiving end $R$. The sending-end source feeds the cable through a disconnector; at the receiving end a second disconnector connects the cable to the load impedance $Z_L$. The voltages $U_S$ and $U_R$ are measured against ground at the two cable ends, the currents $I_S$ and $I_R$ are counted positive in the direction from $S$ to $R$, and $P_S, Q_S$ and $P_R, Q_R$ denote the active and reactive power entering and leaving the cable of length $L_{\text{cable}}$.

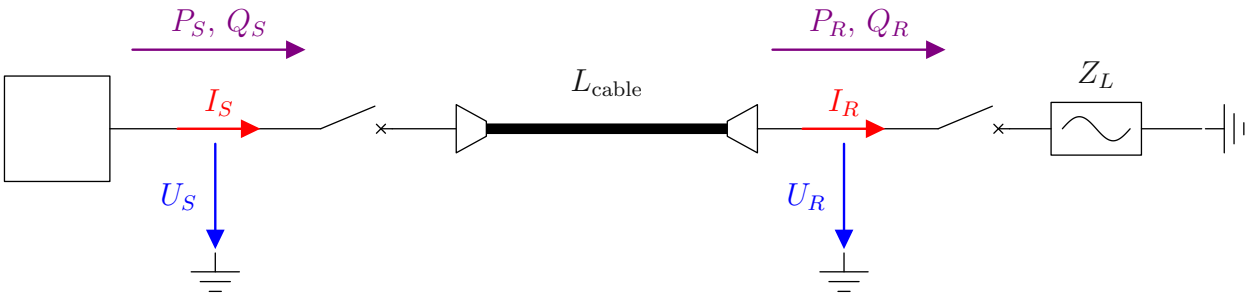



**Figure 1:** Single-line diagram of the cable transmission link (TikZ).

The figure is drawn in TikZ (`figures/transmission-line-tikz.tex`) and can be reused in other documents with `\input`.

## 2 Comparison with the original

Figure 2 reproduces the original drawing (`figures/image9.pdf`) that the TikZ figure was traced from, for a side-by-side check of geometry, colours and labels. Both figures are scaled to the full text width, so they can be compared directly.

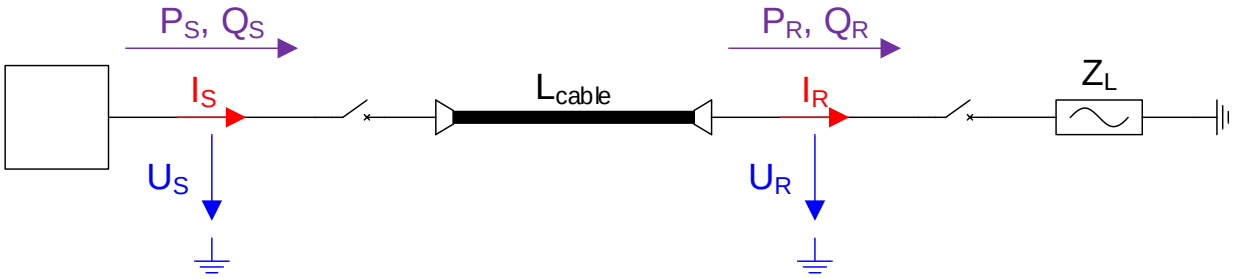



**Figure 2:** Original drawing used as reference.

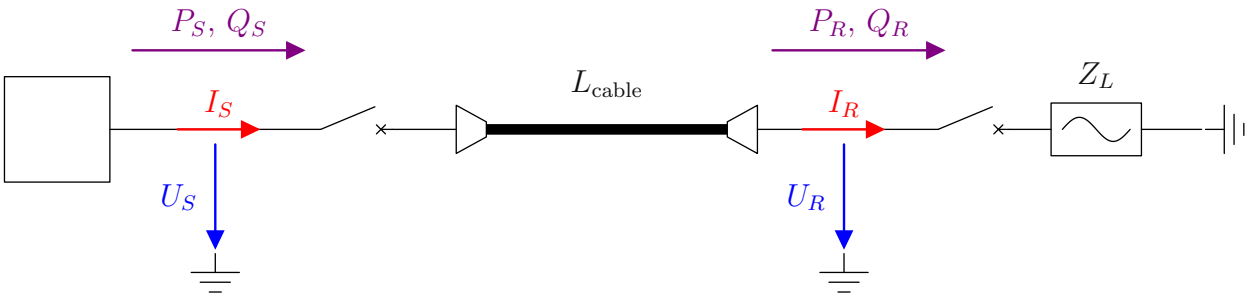



**Figure 3:** TikZ rendering of Figure 2.